

Effective Diameter of Crusher:

$$D_{eff} = D_w + 2 * H/4 = 2 + 2 * (.5)/4 = 2.25$$

$$D_{eff} = D_w + 2 \left(\frac{H}{4} \right)$$

$$= 2 + 2 \left(\frac{0.5}{4} \right)$$

$$= 2.25 \text{ in}$$

Effective Area of Crusher:

$$A_m = \frac{\pi (D_{eff}^2 - d_{hole}^2)}{4}$$

$$= \frac{\pi ((2.25 \text{ in})^2 - (0.5 \text{ in})^2)}{4}$$

$$= 3.780 \text{ in}^2$$

Stiffness:

$$k_m = \frac{A_m E_c}{t_c}$$

$$= \frac{(3.780 \text{ in}^2)(13,050 \text{ psi})}{0.5 \text{ in}}$$

$$= 98,658 \text{ lbf/in}$$

Core Crush Stress:

$$\Delta = \frac{N_T}{N}$$

$$= \frac{(1/2)}{20}$$

$$= 0.025 \text{ in}$$

Bolt Preload:

$$P = k_m \Delta$$

$$= (98,658)(0.025)$$

$$= 2,466.5 \text{ lbf}$$

Effective Diameter Beneath Facesheet:

$$D_{eff2} = D_w + 2t_u$$

$$= 0.5 + 2(0.03)$$

$$= 0.56 \text{ in}$$

Crush Stress:

$$\begin{aligned}f_c &= \frac{P}{\pi \left[\left(\frac{D_{\text{eff2}}}{2} \right)^2 - \left(\frac{d_{\text{hole}}}{2} \right)^2 \right]} \\&= \frac{2466}{\pi \left[\left(\frac{0.56}{2} \right)^2 - \left(\frac{0.5}{2} \right)^2 \right]} \\&= 49,368 \text{ psi}\end{aligned}$$

Max Compressive Stress of the Foam Core: $\sigma_c = 203psi$!!!

Max Preload Possible:

$$\begin{aligned}P_{max} &= \sigma_c A \\A &= \frac{\pi}{4} ((D_w)^2 - (d_{hole})^2) \\A &= \frac{\pi}{4} (2^2 - .5^2) \\A &= 2.945 in^2 \\P_{max} &= (203psi)(2.945 in^2) \\P_{max} &= 597 lbf = 2655 N\end{aligned}$$